

Project Design Phase

Problem – Solution Fit Template

Date	29 June 2026
Project Name	OptiCrop
Maximum Marks	2 Marks

Problem – Solution Fit Template

The **Problem–Solution Fit** helps determine whether the proposed solution effectively addresses the real challenges faced by the target users. It enables the project team to understand customer needs, identify pain points, and validate that the solution provides meaningful value. By aligning user problems with an intelligent, data-driven solution, OptiCrop aims to improve agricultural productivity, optimize resource utilization, and support informed farming decisions.

Purpose

- Understand the major challenges faced by farmers in selecting suitable crops and managing agricultural resources effectively.
- Provide personalized crop recommendations using soil nutrients, weather conditions, rainfall, humidity, temperature, pH, and other environmental factors.
- Help farmers maximize crop yield while minimizing resource wastage through data-driven insights.
- Improve decision-making by delivering accurate, timely, and easy-to-understand recommendations.
- Increase trust and adoption by offering a reliable platform that solves common farming problems with machine learning and predictive analytics.
- Support sustainable agricultural practices by optimizing fertilizer usage, water management, and crop planning.
- Continuously analyze agricultural data to enhance recommendation accuracy and improve the overall farming experience.

Problem–Solution Fit Canvas (OptiCrop Project)

Purpose / Vision: Empower farmers with AI-driven insights for better decisions and higher yields. Version: 1.0

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS Primary Customers: - Small and marginal farmers - Commercial and large-scale farmers - Farmer cooperatives & FPOs Secondary Customers: - Agricultural advisors & agronomists - Agri-input suppliers - Government agricultural extension agencies - Financial institutions & insurers Geography: Rural and semi-urban agricultural regions.	6. CUSTOMER CONSTRAINTS CC - Lack of timely, accurate, and location-specific crop information. - Difficulty in identifying pests, diseases, and nutrient deficiencies early. - Uncertainty in weather, affecting planning and yields. - Limited access to expert advice and field visits. - High input costs and inefficient resource usage (water, fertilizers, pesticides). - Lack of data-driven insights for better decision making.	5. AVAILABLE SOLUTIONS AS - Traditional methods: farmer experience, guesswork. - Local agricultural advisors and extension services. - Government apps and portals (weather, schemes, advisories). - Generic crop management apps. - IoT-based soil and weather monitoring devices. - Satellite-based crop monitoring platforms. Gap: Existing solutions are either not localized, not actionable, not affordable, or not easy to use for everyday farmers.	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P - Know what crop to grow and when. - Monitor crop health and detect problems early. - Get personalized recommendations for inputs and practices. - Monitor weather and receive timely alerts. - Improve productivity and reduce input costs. - Make better decisions to increase profit and reduce risk. Core Problems: - Low productivity and yields. - Crop losses due to pests, diseases, and weather. - Over/under use of water, fertilizers, and pesticides. - Lack of actionable, trusted, and easy-to-understand insights.	9. PROBLEM ROOT CAUSE RC - Information is not localized or timely. - Farmers cannot identify issues in early stages. - Recommendations are generic, not specific to field conditions. - Data is scattered (weather, soil, crop) and not actionable. - Limited digital tools that are user-friendly and in local languages. - High dependency on manual processes and guesswork.	7. BEHAVIOUR BE - Farmers prefer simple, easy-to-use tools. - Rely on experience and word-of-mouth advice. - Prefer actionable recommendations they can trust. - Check mobile phones regularly. - Likely to adopt tools that show clear benefits (yield, savings). - Value solutions in local language with visual guidance.	
3. TRIGGERS TR - Unusual changes in weather. - Visible symptoms on crops or soil issues. - Low yield in previous season. - Advisory from peers or agricultural experts. - Promotions, demo, or training by OptiCrop team.	10. YOUR SOLUTION (OptiCrop) SL OptiCrop – AI-powered crop advisory and monitoring platform. Key Solutions: - AI-based crop health monitoring (image & sensor-based). - Personalized crop advisory (sowing, irrigation, fertilization, pest & disease). - Real-time weather updates and intelligent alerts. - Soil health insights and recommendations. - Easy-to-use mobile app in local languages. - Dashboard for records, analytics, and decision support. How it fits: OptiCrop provides timely, localized, and actionable insights in a simple format, helping farmers make the right decisions, optimize inputs, increase yield, and maximize profits.	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE - OptiCrop Mobile App (Android) - Website Dashboard - WhatsApp / SMS Alerts - Social Media (YouTube, Facebook, Instagram) - Online Ads & Agri Communities 8.2 OFFLINE - Farmer Training & Workshops - Field Demonstrations - Agri Input Stores & Dealer Networks - Local Krishi Melas & Events - Print Materials (Posters, Leaflets, Pamphlets)	Extract online & offline CH of BE	
4. EMOTIONS: BEFORE / AFTER EM Before: - Worried, uncertain, stressed about crop loss and low yield. - Confused about what action to take. After: - Confident, informed, and in control. - Satisfied with higher yield and better profits. - Peace of mind with timely alerts and recommendations.				

Define CS, fit into CC

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

Extract online & offline CH of BE